Process Design Document

LOAN REQUEST INTAKE, CRITERIA VALIDATION and APPROVAL AUTOMATION

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Document History

Date	Version	Role	Name	Organization (Dept.)	Function	Comments
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I. Introduction

I.1 Purpose of the Document

The Process Definition Document outlines the business process chosen for automation using UiPath Robotic Process Automation (RPA) technology.

The document describes the sequence of steps performed as part of the business process, the conditions and rules of the process prior to automation and how they are envisioned to work after automating it, partly or entirely. This specifications document serves as a base for developers, providing them the details required for applying robotic automation to the selected business process.

I.2 Objectives

The process that has been selected for RPA is part of the **Loan Application Management workflow** conducted within the operations of a generic bank. The business objectives and benefits expected by the Business Process Owner after automation of the selected business process are:

- **Reduce the manual processing time of each loan application by up to 80%** through automated data validation and entry.
- **Increase accuracy and compliance** by validating every application against predefined business rules before submission to the bank's internal system.
- **Ensure traceability and better monitoring** of the entire loan application process through the logs and audit trails generated by the automation.
- **Improve customer experience** by sending immediate confirmation emails to applicants whose loan requests have been approved.

I.3 Key Contacts

The specifications document includes concise and complete requirements of the business process and it is built based on the inputs provided by the process **Subject Matter Expert (SME)/ Process Owner.**

The **Process Owner** is expected **to review it and provide signoff for accuracy** and completion of the steps, context, impact and complete set of process exceptions. The names have to be included in the table below.

Role	Name	Contact details (email, phone number)	Notes
Process SME	Aris-Georgian ILIE	arisgeorgian2004@gmail.com	Point of contact for questions related to process details & exceptions
Process Reviewer	Aris-Georgian ILIE	arisgeorgian2004@gmail.com	Point of contact for questions related to process details & exceptions

Process Owner/	Aris-Georgian ILIE	arisgeorgian2004@gmail.com	Escalations, Delays etc.

I.4 Minimum Prerequisites for Automation

1. **Filled in Process Design Document**: a complete and approved Process Design Document (PDD) is required as the baseline for development.
2. **Test Data to Support Development**: the test data for this process is provided directly from the loan application form created in **Google Forms**. The responses are automatically stored in a corresponding **Google Sheet**, which serves as the input source for the automation. The automation retrieves this data for validation and further processing during development and testing.
3. **User Access and User Accounts Creation**: the necessary **UiPath environment** must be available, including UiPath Studio. User accounts, licenses and permissions are required to create and run robots.
4. **Credentials**: are required for secure access to the machines and applications. This includes authentication for **Gmail, Google Drive and Google Sheets**. Credentials should be securely stored and managed.
5. **Dependencies with Other Projects in the Same Environment**: the automation requires connections and integrations to the applications used: **Mail, Gmail, Google Drive, Google Sheets**. Additionally, it is necessary to have **Excel input files** available.

II. As-Is Process Description

II.1 Process Overview

General information about the process selected for RPA prior to automation.

#	Item	Description
1	**Process full name**	Loan Request Intake, Criteria Validation and Approval Automation
2	**Process Area**	Banking Operations
3	**Department**	Loan Management
4	**Process short description** (operation, activity, outcome)	Applicants manually submit their loan requests using a form (on paper or as a basic digital file). A loan officer collects all the forms and manually enters the data into an Excel file at the end of the day. Once completed, the officer attaches the Excel file to an email and sends it to another colleague responsible for processing loan applications. The receiving officer downloads the file from the email, opens it in Excel and reviews each loan request against the bank's eligibility criteria (e.g., applicant's age, declared income, requested loan amount). For each request, the officer manually enters the validated data into the UiBank system. The officer records the generated Loan ID and then manually sends a confirmation email to the applicant, informing them that their loan has been approved.

5	**Role(s) required for performing the process**	Loan Officers
6	**Process schedule and frequency**	Daily, Monday to Friday, 9 am - 6 pm
7	**# of items processes /reference period**	~90 loan requests / day (based on average intake from applicants)
8	**Average handling time per item**	4 - 6 minutes per application (manual data entry, validation, system update and communication)
9	**Peak period (s)**	Beginning of the month, after salary payments, when more applicants request loans
10	**Transaction Volume During Peak period**	100 - 150 loan requests / day
11	**Total # of FTEs supporting this activity**	3 - 5 Loan Officers
12	**Expected increase of volume in the next reference period**	Loan application volumes expected to increase by ~15-20% due to higher demand and seasonal trends
13	**Level of exception rate**	7 - 12% (invalid, incomplete, or non-eligible applications rejected at validation stage)
14	**Input data**	Loan application forms (paper or digital files) collected from applicants; data manually entered into Excel by loan officers
15	**Output data**	Loan IDs generated in UiBank recorded manually; confirmation emails written and sent manually to eligible applicants

II.2. Applications Used in the Process

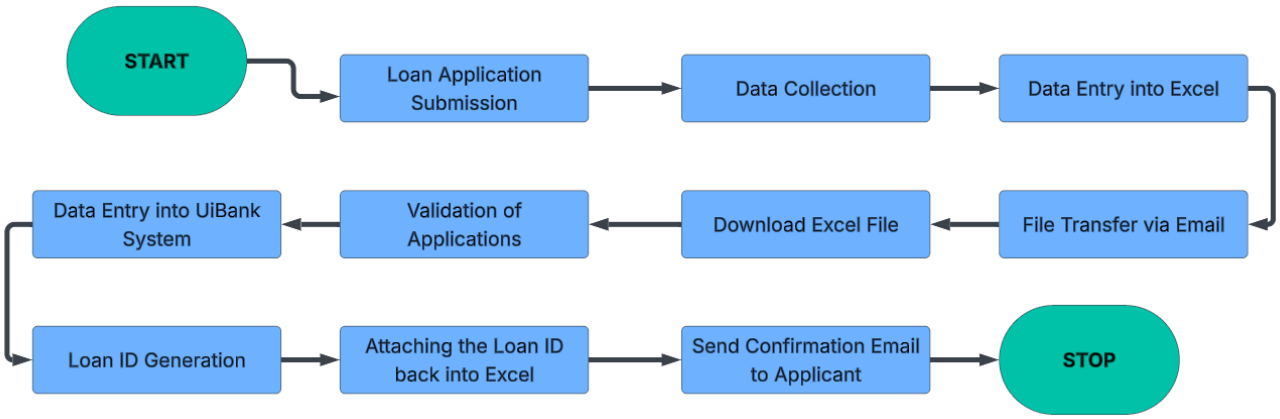
The table includes a comprehensive list all the applications that are used as part of the process automated, at various steps in the flow.

#	Application name & version	System Language	Thin/Thick Client	Environment/ Access method	Comments
1	Gmail	EN	Thin/Thick Client	Web-based access via browser or desktop client	Used to send and receive the Excel file containing the loan requests between staff members.
2	UiBank	EN	Thin Client	Web Browser	Used by loan officers to manually input loan application details, check eligibility and generate Loan IDs.
3	Excel	EN	Thick Client	Windows Application	Used by loan officers to manually record loan applications, validate details and prepare files for further processing.
4	Manual Email Client	EN	Thin/Thick Client	Web Browser or Desktop Client	Used by staff to manually compose and send confirmation emails to applicants whose loans were approved.

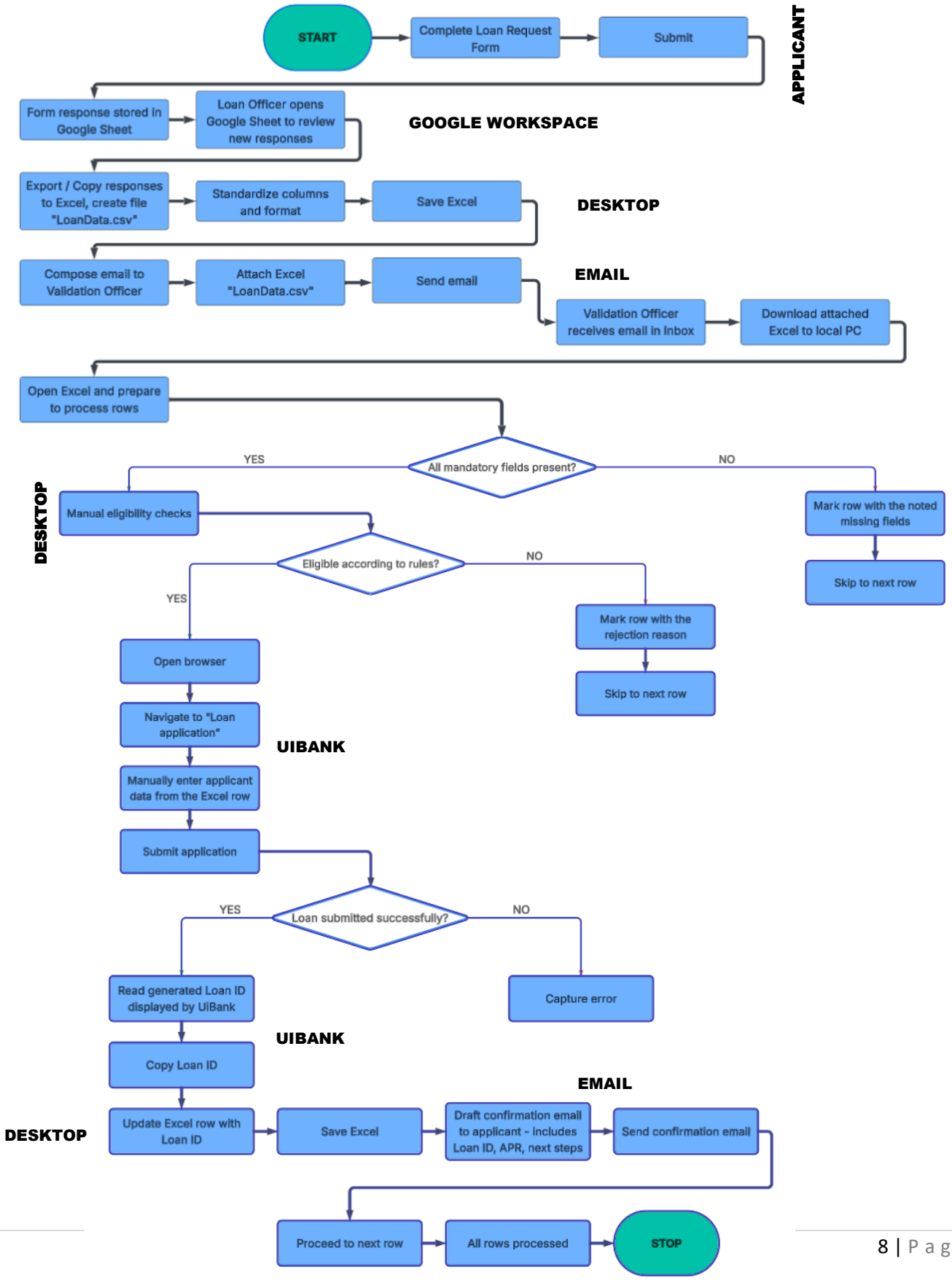
II.3 As-Is Process Map

High Level As-Is Process Map:

This chapter depicts the As Is business process at a High Level to enable developers to have a high-level understanding of the current process.



Detailed As-Is Process Map:



II.4 Detailed As-Is Process Steps

This chapter depicts the As-Is business process in detail to enable the Developer to build the automated process.

Check the attached document.

Process_Details.xlsx

II.5 Input Data Description

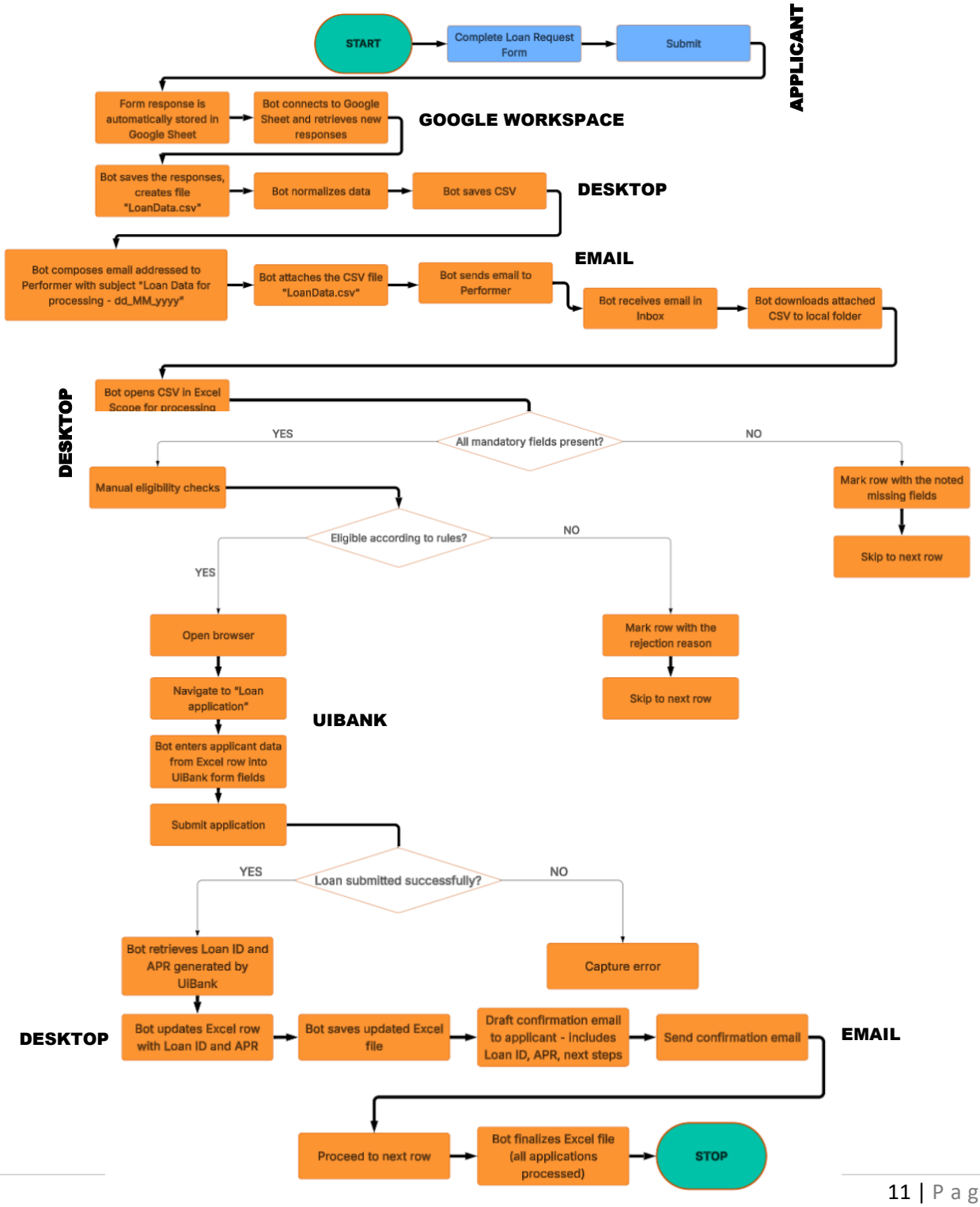
Step	Input type	Location	Inputs are standard? (Yes/ NO)	Inputs are structured?	Data to be used from
1	Web Form	Google Forms (submitted by applicant)	YES - all applicants use the same form template	YES - fields are structured and machine-readable	Applicant Name, Age, Email Address, Annual Income, Requested Loan Amount, Loan Term
2	Spreadsheet	Google Sheets (linked automatically to Google Form)	YES - responses always appear in fixed columns	YES - structured (digital tabular format)	All form fields
4	CSV File	Local PC / Google Drive	YES - file structure is consistent with columns	YES - structured tabular format	Same as Google Sheet fields + additional Comment columns for processing
11	Spreadsheet Row	Excel File (Validation Officer view)	YES - fixed structure per row	YES - structured	Applicant details row (mandatory fields check: Name, Email, Age, Income, Loan Amount, Term)
13	Spreadsheet Row	Excel File	YES	YES	Same row as above + Business Rule check results (Rejection Reasons)

17	Web Form	UiBank Web Application	YES - UiBank form layout is fixed	YES - structured input fields	Applicant data manually entered from Excel (Email, Age, Income, Loan Amount, Term)
20	Screen Data	UiBank Confirmation Page	YES - Loan ID/Approval always in fixed position	YES - digital and machine-readable	Loan ID, APR
22	Spreadsheet Row	Excel File	YES - fixed columns	YES - structured	Applicant details + Loan ID + APR + Status
25	Email	Outlook/Gmail	YES - template used	YES - structured (digital text with placeholders)	Loan ID, APR, Applicant Name, Email address

III. To-Be Process Description

This chapter highlights the expected design of the business process after automation.

III.1 To-Be Detailed Process Map



III.2 Parallel Initiatives/ Overlap (if applicable)

This chapter captures the proposed Business, Process & System changes in near future and its impact

S. No	Initiative Name	Process Step(s) where it is identified	Impact on current automation request? How?	Expected Completion Date	Contact person for more details
1	Google Forms structure update	Applicant Form submission (Step + Bot retrieval from Google Sheets	If the Google Form is modified (new fields, removed fields), the bot's mapping to mandatory fields will break. Requires updates to field extraction logic.	TBD (depends on business team managing the form)	Business Analyst / Loan Intake Owner
2	Google Workspace API changes	Dispatcher Bot	If Google Sheets API or Gmail API changes, data extraction or email sending may fail. UiPath integrations may need to be updated.	According to Google Workspace release schedule	System Administrator (Google Workspace)
3	UiBank upgrades	UiBank steps	Any changes in UiBank UI (new fields, page redesign, login flow) will impact selectors and form filling logic. Bot will require selector adjustments.	TBD (UiPath may update UiBank periodically)	UiPath Training/Support Team
4	Business rules change (eligibility criteria)	Performer Bot validations	If income thresholds, loan limits, or age requirements change, the rules coded in the Performer must be updated.	As per future policy updates	Business Process Owner (Loan Validation)
5	Compliance / Data Privacy policy updates	Data handling (Excel saving, Email sending, E8)	Stricter data storage or encryption requirements may affect how CSV/Excel is saved and how emails with personal data are sent.	TBD (regulatory-driven)	Compliance Officer

III.3 In Scope for RPA

The activities **in scope of RPA**, are listed here:

1. Capture loan request responses from Google Sheets
2. Save loan request data into a standardized CSV file
3. Normalize and format data
4. Save CSV file
5. Compose and send email with attached CSV to Performer
6. Download CSV attachment from email
7. Open and process CSV file in Excel scope
8. Validate mandatory fields (Name, Age, Email, Income, Loan Amount, Loan Term)
9. Mark incomplete applications in Excel with reasons for disqualification
10. Apply eligibility criteria rules (Age, Income threshold, Loan Amount, Loan Term validity, Email presence)
11. Mark rejected applicants in Excel with rejection reasons
12. Navigate to the Loan Application page in UiBank
13. Input applicant data from Excel into UiBank loan application form
14. Submit loan application in UiBank
15. Retrieve Loan ID and APR generated by UiBank
16. Update Excel row with Loan ID
17. Save updated Excel file
18. Generate confirmation email using predefined template
19. Send confirmation email to approved applicants
20. Loop through all rows until all applications are processed

III.4 Out of Scope for RPA

The activities **OUT of scope of RPA**, are listed here:

Sub-process	Activity (step)	Reasons for Out of Scope	Impact on the To-Be	Possible measures to be taken into consideration for future automation
1	Applicant fills in Loan Request Form - Google Form submission	External action by customer; cannot be automated because it requires applicant input	Remains a manual step, but triggers the automated flow	Could be extended in the future with integrations from other channels (mobile app, chatbot) feeding into Google Forms/Sheets automatically

III.5 Business Exceptions Handling

The Business Process Owner and Business Analysts are expected to document below all the business exceptions identified in the automation process. These can be classified as:

Known	Unknown
Previously encountered. A scenario is defined with clear actions and workarounds for each case.	New situation never encountered before. It can be caused by external factors. Cannot be predicted with precision, however if it occurs, it must be communicated to an authorized person for evaluation.

Known Exceptions

The table below reflects all the business process exceptions captured during the process evaluation and documentation. These are **known exceptions,** met in practice before. For each of these exceptions, define a corresponding expected action that the robot should complete if it encounters the exception.

BE #	Exception name	Step	Parameters	Action to be taken
1	No Email provided by applicant	Validation step - Check mandatory fields	Email field missing	Bot marks incomplete applications in Excel with reasons for disqualification, skip to next
2	Email received without attachment	File reception step	Missing CSV attachment in email	Bot sends notification to business team, stops batch processing
3	CSV file has no data	File processing step	Empty CSV file	Bot terminates run, logs warning
4	Invalid Loan Term - not 1, 3, 5, 10 years	Eligibility validation step - Apply business rules	Loan Term value	Bot marks incomplete applications in Excel with reasons for disqualification, skip to next

5	Applicant Age < 18	Eligibility validation step - Apply business rules	Age field	Bot marks incomplete applications in Excel with reasons for disqualification, skip to next
6	Loan Amount ≥ 3 * Yearly Income	Eligibility validation step - Apply business rules	Loan Amount, Yearly Income	Bot marks incomplete applications in Excel with reasons for disqualification, skip to next
7	Loan Amount ≥ 100,000	Eligibility validation step - Apply business rules	Loan Amount	Bot marks incomplete applications in Excel with reasons for disqualification, skip to next

Unknown Exceptions

For all the other **unanticipated or unknown business (process) exceptions**, the robot should:

1. Stop processing the current transaction and log the exception details
2. Continue with the next transaction, unless the error prevents further processing of the batch
3. Escalate the case to an authorized person for evaluation and resolution

III.6 Application Error and Exception Handling

A comprehensive list of all errors, warnings or notifications should be consolidated here with the description and action to be taken, for each, by the Robot.

Errors identified in the automation process can be classified as:

Area	Known	Unknown
Technology/ Applications	Experienced previously, action plan or workaround available for it.	New situation never encountered before or may happened independent of the applications used in the process.

Know Errors or Exceptions

The table below reflects all the errors identifiable in the process evaluation and documentation.

For each of these errors or exceptions, define a corresponding expected action that the robot should complete if it is encountered.

#	Error name	Step	Parameters	Action to be taken
1	Application Crash / Internal Server Error	Any step	Error message	Recover & retry up to 3 times; if failure persists, close applications and restart
2	Timeout while loading UiBank page	UiBank navigation	Timeout exception	Retry navigation up to 2 times; if still failing, mark transaction
3	Invalid Selector or Element not found	UiBank data entry	Selector not found	Retry selector resolution; if still failing, mark row

Unknow Errors and Exceptions

For all the other **unanticipated or unknown application exceptions/errors**, the robot should:

1. Log the error details
2. Mark the affected transaction
3. Notify the business owner or support team
4. Retry the action if possible
5. Escalate the case to an authorized person if the issue persists
6. Continue with the next transaction if the error is isolated